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ORIGINAL RESEARCH ARTICLE

The feasibility of voluntary counselling and HIV testing for pregnant women using community volunteers in Zimbabwe

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Summary: The purpose of this pilot project was to assess the feasibility and acceptability of voluntary counselling and HIV testing (VCT) by pregnant women using community volunteers in Zimbabwe to prevent mother to child transmission (MTCT) of HIV. From July 1999 to June 2001, a short-course zidovudine (ZDV)-based perinatal HIV prevention programme was initiated in two antenatal clinics. Community volunteers, recruited from local community organizations, underwent a two-week training course in VCT, which included HIV/AIDS facts, systematic counselling approach, and practical counselling techniques using scripts and role-play. Rapid HIV testing was performed after informed consent. Lay counsellors conducted individual pre- and post-test counselling for HIV. A total of 35 women community volunteers were trained in VCT; 34 graduated and committed to work four hours per week in the clinic. Of the 6051 pregnant women presenting for antenatal clinics (ANC), 1824 (30%) underwent pre-test counselling and 1547 (26%) were tested, and 429 (28%) were HIV infected. Overall, 1283 (83%) returned for their test results including 406 (95%) of HIV-infected women. Of the 406 HIV-infected women who collected their test results, only 203 (50%) opted for ZDV prophylaxis to prevent MTCT of HIV. Over the two-year study period, two counsellors died and three sought employment at other organizations. Adherence to duty roster was 97% and no breach of confidentiality was reported. Despite many challenges, VCT delivered by community volunteers is feasible and acceptable for pregnant women aiming to reduce their risk of transmitting HIV to their infants. This programme is being implemented at several urban and rural MTCT sites in Zimbabwe and can serve as a model for other resource-poor countries.

Keywords: HIV, VCT, MTCT, lay community volunteers, Zimbabwe

Introduction

Zimbabwe has one of the highest prevalence of HIV infection worldwide; approximately 25–30% of the adult populations are estimated to be infected, primarily through heterosexual contact.[1–3] Recent estimates indicate that more than 30% of pregnant women attending antenatal clinics (ANC) in the capital city of Harare are HIV seropositive.[3] Without any intervention, a third of these women will give birth to infected babies; 50,000 infants acquire infection from their mothers annually in Zimbabwe.[4] In recent years, several clinical trials have demonstrated the efficacy of simpler and less expensive short courses of: zidovudine (ZDV); ZDV/lamivudine (3TC); or single-dose nevirapine (NVP) in reducing mother to child transmission (MTCT) of HIV in sub Saharan Africa.[5–7]

Voluntary counselling and HIV testing (VCT) during prenatal care offers the best opportunity to access these antiretroviral prophylactic regimens.[8] However, the high cost of VCT raises concern about the affordability of VCT in resource-poor settings.[9] Clinics and hospitals are severely understaffed with nurses, who traditionally are the primary care-givers in many periurban clinics in Zimbabwe. Nurses are often trained in counselling, but the increased clinic workload due to staff shortage has resulted in a reluctance to take on the additional task of counselling. In addition, there are no

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available funds to employ and train a new cadre of full-time professional counsellors to deliver VCT. For these reasons, we conducted a pilot study to assess the feasibility and acceptability of VCT among pregnant women using community volunteers at antenatal clinics in urban Zimbabwe.

Methods

Study population, design and data collection

From July 1999 to June 2001, an open-label prospective cohort study of a maternal short-course ZDV regimen to prevent MTCT of HIV (sponsored by the Swedish International Development Agency) was initiated at Seke North and St. Mary antenatal clinics in Chitungwiza, Zimbabwe, which is a periurban community 25 km south of the capital city of Harare, where approximately 2400 deliveries occur annually. Rapid HIV testing was performed after informed consent. Institutional review board and human subjects committees at Stanford University School of Medicine, Palo Alto, CA, USA, and at the Medical Research Council of Zimbabwe approved the study.

Thirty-five community volunteers were recruited from Red Cross community organization. The Red Cross has a long tradition of volunteers and currently implements the home-based care programme in Chitungwiza. Churchwomen who attend the 10 churches in the clinic catchment area were approached. Eligible counsellors were 25 years of age or older, willing to volunteer, and willing to commit to at least one counselling session per week for a period of 36 weeks. Incentives included a certificate for completion of the training course, refreshments during training, formal public acknowledgement at their church, or organization during meetings and a badge. The project counselling coordinator and the community liaison officer conducted a two-week training workshop in VCT at the community hall. The materials used in training included HIV/AIDS facts, systematic counselling approach, and practical counselling techniques using scripts and role-play. Community volunteers conducted individual pre- and post-test counselling.

Study participants included women who booked for antenatal care at the two antenatal clinics were eligible for the ZDV intervention trial due to late gestational age (>34 weeks by fundal height) were offered inclusion in this study. Informed consent was obtained from each participant.

HIV testing

Maternal HIV status was determined on site using two rapid tests (Capillus Test, a latex agglutination assay – Cambridge Diagnostics Ireland Limited, Galway, Ireland and Dipstick Test, a dot immunoassay – Immuno Chemical Laboratory, Bangkok, Thailand) on each blood sample. If the two rapid tests were positive, an enzyme-linked immunoassay (EIA, Organon Teknika Corporation, Durham, NC, USA) was performed on a new blood sample to confirm HIV infection. Western blot (Organon Teknika Corporation, Durham, NC, USA) was performed for two samples from the same blood collection only when the two rapid tests yielded ambiguous or discordant test results. In accordance with the recommendations of the National AIDS Control Policy, a woman was considered to be HIV positive if she had two positive rapid HIV tests confirmed by an EIA or a positive Western blot test if the rapid test results were ambiguous or discordant. HIV-negative test results (both tests negative) were offered to clients the same day. HIV-positive test results were available to clients after one week since the EIA was done at the reference laboratory.

Results

A total of 35 women community volunteers were trained in VCT; 34 graduated and committed to work four hours per week in the clinic. Community volunteers conducted individual pre- and post-test counselling. Of the 6051 pregnant women presenting for their first antenatal care visit, 1824 (30%) underwent pre-test counselling for HIV and 1547 (26%) were tested, and 429 (28%) were found to be HIV infected. Overall, 1283 (83%) returned for their test results including 877 (78%) HIV-uninfected women and 406 (95%) HIV-infected women (Figure 1). Of the 406 HIV-infected women who returned to collect their test results and were informed about the role of ZDV prophylaxis to prevent MTCT of HIV during post-test counselling, only 203 HIV-infected women (50%) started ZDV prophylaxis and 177 completed prophylactic treatment; 26 women dropped out of the study for a variety of reasons including premature deliveries, lost to follow-up, social issues at home preventing participation, and low maternal haemoglobin (Hgb $<$ 8 g/dL).

Of the 1547 women who underwent VCT, only 93 male partners (17%) agreed to get tested for HIV. The 203 HIV-infected women who enrolled in the project joined a support group for HIV-infected individuals in Chitungwiza, and a Community Advisory Board (CAB) was formed. In addition, a Community Volunteer Counsellor Training Manual was formulated, and networking and linkage with other HIV/AIDS organizations were established. Over the two-year study period, two counsellors died and three sought employment at other organizations; all five counsellors were subsequently replaced. Adherence to duty roster was 97% and no breach of confidentiality was reported. The entire counselling staff was retained for the entire project period of two years.

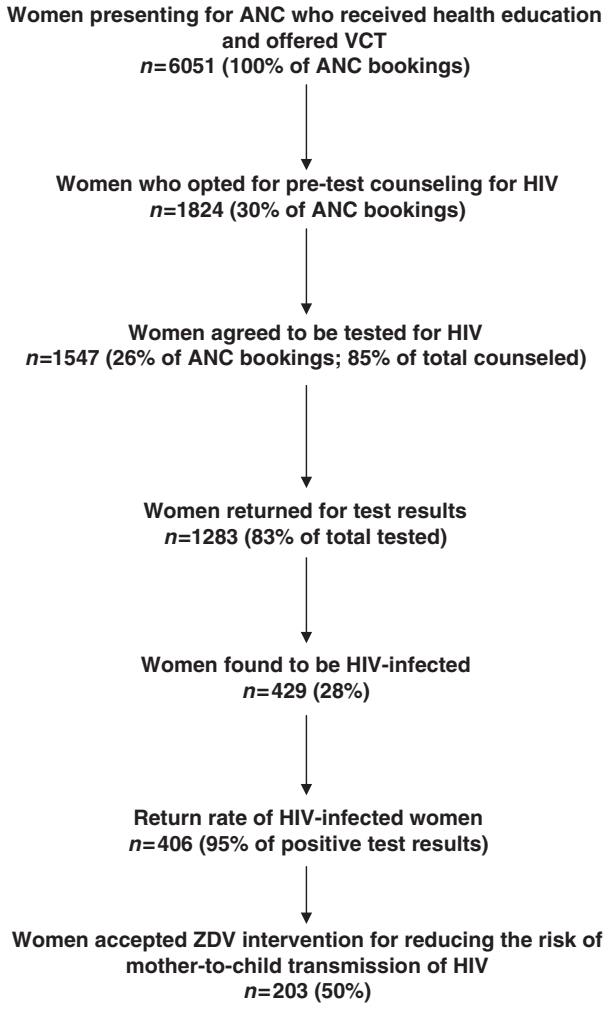


Figure 1 Acceptability of voluntary counselling and HIV testing using community volunteers in Zimbabwe

Discussion

The primary aim of this pilot project was to assess the feasibility and acceptability of lay community volunteer-assisted VCT among pregnant women in the context of SIDA sponsored MTCT prevention programme using short-course ZDV regimen in a periurban antenatal clinic setting in urban Zimbabwe. In Zimbabwe, economic hardships and political instability have taken the toll on health services.[10] Although nurses often are trained in VCT, there currently is inadequate staff at clinics and hospital settings to perform counselling in addition to the other competing patient care duties. In such circumstances, use of lay counsellors, either paid or volunteer, can augment counselling capacity.[8] Our experience suggests that pre- and post-test HIV counselling in the ANC setting by community volunteers is feasible, effective, and acceptable by both clinic staff and clients. Similar experiences have been reported from other countries in sub-Saharan Africa.[11–13]

The prevalence of HIV infection in Zimbabwe is one of the highest in the world. In the present study, 28% of women were HIV positive; this finding is consistent with other antenatal clinic population point prevalence data that have been reported in Zimbabwe.[3] The finding that only 30% of pregnant women opted for HIV counselling is of great concern. Entrance into counselling is the major barrier, but once counselled 85% of these women opted to be tested for HIV. There are differences in attitude towards VCT depending on the community, and medical and social support available for HIV-infected women.[14] Studies conducted in North America and Europe show that women who perceive themselves as at risk of HIV are more likely to accept VCT,[15,16] since in these settings the benefits of testing is perceived to be high such as access to prenatal care, availability of formula feeding, and easy access to antiretroviral therapy (ARV) for preventing MTCT as well as highly active ARV for the mother.[17] In contrast, a number of social and health service barriers to VCT exist in resource-poor settings.[8,14,18] In recently conducted focus group discussions with our study participants, barriers to VCT included fear of domestic violence and desertion upon disclosure to partner and in-laws, denial of a positive diagnosis, lack of availability of ARV, and stigma.[19] These factors have been identified in other resource-poor settings as well.[18,20,21]

Due to the stigma associated with HIV/AIDS, and the widespread fear of testing, innovative approaches to increase access and uptake of VCT among pregnant women are urgently needed.[8,14] Another ZDV-based perinatal HIV prevention trial conducted in West Africa have reported a low return rate for HIV-positive test results.[21] In our study, the rate of collection of positive test results was 95% even though test results were available to women one week after they were tested for HIV. Use of rapid on-site HIV testing with same-day availability of test results is an effective strategy in MTCT programmes to improve VCT uptake rates,[22,23] and should be expanded to include couples to leverage a decrease in heterosexual transmission as well.[22,23] Other strategies to enhance delivery of proven ARV prophylactic interventions to prevent MTCT of HIV could include group pre-test counselling,[11] universal offer of testing with women having the option to 'opt out', universal treatment (mass treatment for those with undetermined HIV status by VCT), universal testing with women having the right to 'opt out' from learning test results or mass treatment for all without testing.[8] Recent data from the USA suggest that jurisdictions that shifted from an opt-in approach to either an opt-out or mandatory newborn testing approach have achieved an increase in prenatal HIV-testing rates.[24] Finally, community mobilization and education programmes to promote VCT, which

involves men, seems to be a prudent long-term solution.[8]

Although, pilot projects to prevent MTCT have been initiated in many African countries with the support of United Nations (UN) agencies[25,26] and the Elizabeth Glazer Pediatric AIDS Foundation,[27] uptake of services has been low.[28,29] Only 20% of all pregnant HIV-infected women identified through VCT accept antiretroviral drugs to prevent MTCT.[18,29–32] In our study, only 50% of all HIV-infected women who returned for their test results and were informed of the ZDV-based MTCT programme opted for this intervention.

Despite all the challenges and obstacles we encountered in MTCT programme implementation in a human resource constrained environment, we found that lay community counsellors coped well with the training programme and delivered an acceptable VCT service. Training for a few weeks, followed by ongoing supervision and mentoring by a full-time professional counsellor has resulted in reliable counselling services. In addition to individual pre- and post-test counselling for HIV, the community counsellors provided ongoing counselling and support to HIV-infected women, which included a variety of issues such as bereavement, domestic violence, spousal abandonment, discordant test results, family planning, and negotiating safe sex. The community counsellors also facilitated recruitment of HIV-infected women to local support groups and referred clients for psychosocial support services. The use of community counsellors can augment community support for MTCT prevention programme and can serve as a model for other resource-poor countries. Based on our experience, the ministry of health is currently engaged in implementing VCT services using trained primary care counsellors at several urban and rural MTCT sites in Zimbabwe.

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